Early TEFCA Participation Was Not Associated With Lower Hospital Mortality

A Cross-sectional analysis of 691 U.S. hospitals

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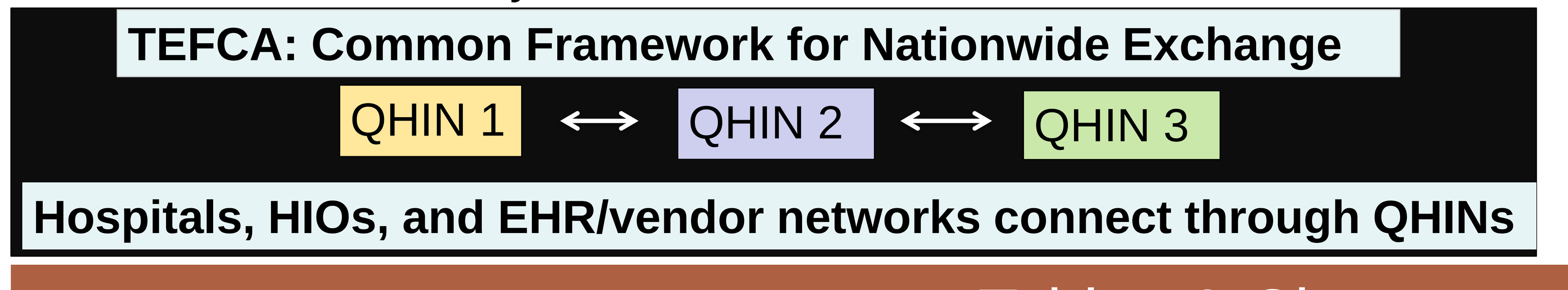
Code, methods, and materials

Background & Objective

The Trusted Exchange Framework and Common Agreement (TEFCA) was established to support nationwide electronic health information exchange (HIE) through a network-of-networks of Qualified Health Information Networks (QHINs).[1] Prior HIE research has reported operational and utilization benefits, but evidence for mortality remains limited and largely observational, particularly during TEFCA's early implementation.[2–11]

Objective: Examine whether current TEFCA participation, compared with planned participation, was associated with lower CMS Hybrid Hospital-Wide Mortality (Hybrid HWM) scores among U.S. non-federal acute care hospitals. Heart failure and pneumonia mortality were evaluated as secondary outcomes.



Methods

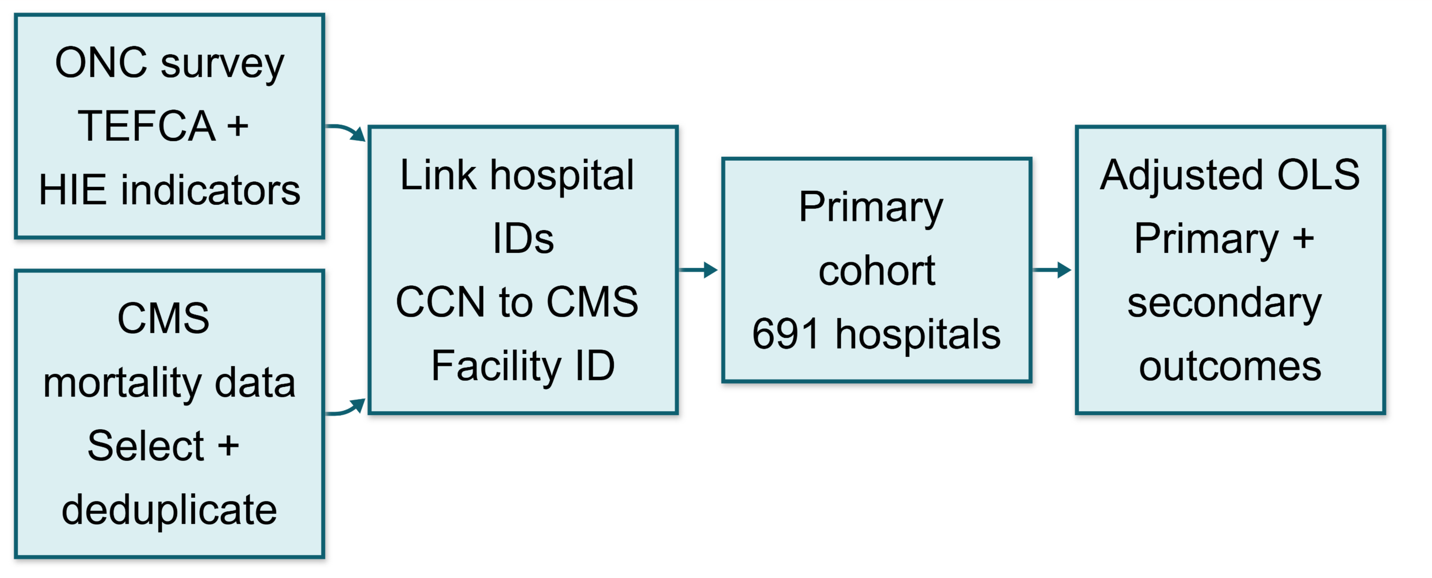


Figure 1. Data linkage and analytic workflow.

Hospital-level ONC survey data (2023–2024)[12] were linked to CMS Hybrid HWM data (7/1/2023–6/30/2024) and heart failure/pneumonia mortality data (7/1/2021–6/30/2024).[13] Adjusted OLS models compared current and neither status with planned participation, controlling for national network, EHR vendor network, HIO participation, log CMS denominator, state, and survey year

Note. Missing TEFCA and other network-participation responses were coded as nonparticipation.

Results

Primary analysis. Among 691 hospitals, 112 (16.2%) reported current TEFCA participation, 289 (41.8%) planned participation, and 290 (42.0%) neither current nor planned. In the adjusted primary model, current participation was associated with a 0.108-point higher Hybrid HWM score than planned participation (95% CI, 0.001–0.216; *p* = .047). The estimate, therefore, did not support lower mortality with current participation.

Robustness and secondary outcomes. With HC3 robust standard errors, the confidence interval included zero (95% CI, −0.010–0.227; *p* = .072). Current-versus-planned estimates were also not significant for heart failure (β = −0.024; *p* = .927) or pneumonia mortality (β = 0.479; *p* = .114). EHR vendor-network participation and larger CMS denominators were associated with lower Hybrid HWM scores.

Key finding: The analyses did not provide robust evidence that early current TEFCA participation was associated with lower hospital mortality than planned participation.

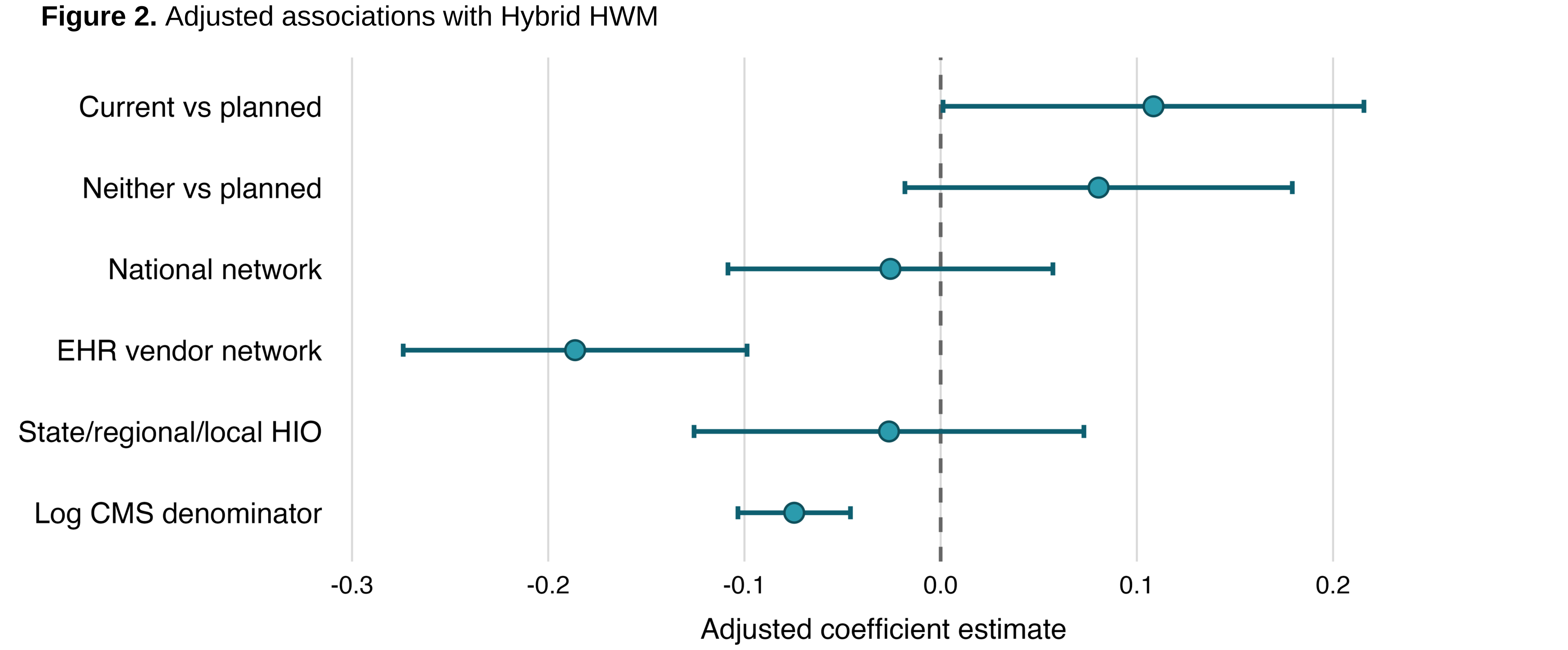
Tables & Charts

Table 1. TEFCA Participation Estimates Across Mortality Analyses

Analysis: current vs planned TEFCA hospitals	β	95% CI	P-value
Primary Hybrid HWM model	0.108	0.001 to 0.216	.047
Primary model, HC3 robust SEs	0.108	−0.010 to 0.227	.072
Heart failure mortality	−0.024	−0.545 to 0.497	.927
Pneumonia mortality	0.479	−0.115 to 1.073	.114

Note. Positive coefficients indicate higher mortality values; planned TEFCA participation was the reference category.

Figure 2. Adjusted associations with Hybrid HWM



Note. Points represent adjusted OLS coefficients; error bars represent 95% confidence intervals. Planned TEFCA participation was the reference category.

Interpretations & Limitations

Interpretation. Early participation in TEFCA was not robustly associated with lower mortality than planned participation. The small positive estimate should not be interpreted as evidence of harm: its statistical significance was sensitive to the standard-error method, and it was not reproduced for secondary outcomes. Clinical effects may require more time, fuller adoption, and routine exchange use to emerge.

EHR vendor-network participation was associated with lower Hybrid HWM, possibly reflecting more established exchange infrastructure or workflow integration, but this secondary finding does not establish causality.

Limitations.

- Cross-sectional hospital-level associations do not establish causality or within-hospital change.
- Survey responses captured reported participation—not exchange volume, duration, data quality, or workflow integration.
- Planned hospitals may already use other exchange networks, narrowing the contrast.
- Unmeasured hospital characteristics may confound estimates.
- The CMS primary outcome window was not wholly post-TEFCA, and the secondary outcome window largely predated TEFCA implementation.

Future work. Longitudinal studies should evaluate sustained TEFCA use, actual exchange activity, and subsequent clinical outcomes.

Conclusion

In this early hospital-level analysis, current TEFCA participation was not robustly associated with lower mortality than planned participation, and the small positive Hybrid HWM estimate was not reproduced across secondary outcomes. Because planned hospitals could already use other exchange networks, this was not a comparison with no electronic exchange. EHR vendor-network participation was associated with lower Hybrid HWM, but this observational finding does not establish causality. Longitudinal studies should evaluate sustained TEFCA use and actual exchange activity as implementation matures.

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